

Notes: Avenancio-León and Howard (QJE, 2022)

The Assessment Gap: Racial Inequalities in Property Taxation

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1 Big picture

- **Question.** Do Black and Hispanic homeowners face higher residential property tax burdens than comparable white homeowners, and if so, through what mechanisms does this inequality arise?
- **Setting.** The paper studies residential property taxation across the United States using property-level assessment, transaction, geography, and race/ethnicity data. The empirical object is the **assessment ratio**: assessed value divided by market value.
- **Tax-equity benchmark.** Within a taxing jurisdiction, all homeowners face the same overlapping tax authorities, public services, intended tax rate, and assessment practices. Under an equitable ad valorem property tax, the assessment ratio should be constant within that jurisdiction.
- **Core finding.** Holding taxing jurisdiction and year fixed, Black homeowners have assessment ratios about **12.7%** higher than non-Hispanic white homeowners; Black or Hispanic homeowners have assessment ratios about **9.8%** higher.
- **Tax-burden implication.** Because tax bills are computed from assessed values, the assessment gap translates into higher effective property tax rates for minority homeowners receiving the same bundle of public services. The paper estimates roughly **\$300–\$390** in additional annual taxes for the median minority homeowner.
- **Spatial decomposition.** Just over half of the overall gap is between neighborhoods. The remaining gap persists within census tracts and even within census block groups.
- **Main mechanism.** Assessments are much less sensitive than market prices to neighborhood characteristics. If assessors underweight neighborhood amenities and disamenities, high-minority neighborhoods can be systematically overassessed relative to market prices.
- **What does not explain the gap.** The assessment gap is not driven by minority homeowners selling at lower transaction prices, nor by racial differences in observable housing-stock attributes such as home size, age, bathrooms, fireplaces, patios, pools, or number of stories.
- **Administrative mechanism.** In Cook County, Black and Hispanic homeowners are less likely to appeal assessments, less likely to win conditional on appealing, and receive smaller reductions conditional on success.
- **Economic takeaway.** A formally race-neutral administrative system can generate racial inequality when valuation errors fall along spatial and bureaucratic lines that are correlated with race and ethnicity because of residential segregation and unequal access to tax-administration processes.

2 Data and measurement (key objects)

2.1 Institutional setting: local property taxation

- Most U.S. local governments levy annual residential property taxes.
- A homeowner’s tax bill is generally computed as:

$$\text{Tax bill}_{ijt} = \tau_{jt} \times A_{ijt},$$

where τ_{jt} is the local policy tax rate and A_{ijt} is the assessed value.

- The assessed value is an administrative estimate of the home’s market value.
- In a pure ad valorem system, equity requires the ratio A/M to be constant for all homes facing the same intended tax burden.
- The paper focuses on the **assessment ratio**, rather than the tax bill itself, because tax exemptions vary by household and can confound comparisons of effective tax rates.

2.2 Taxing jurisdictions

- A **taxing jurisdiction** is defined as a set of homes facing the same overlapping network of public entities: state, county, municipality, school district, and special districts.
- This is more precise than using counties alone, because two homes in the same county can face different school districts or special-purpose levies.
- The authors use GIS shapefiles for more than **75,000** public entities to assign homes to unique taxing jurisdictions.
- Comparing homes within jurisdiction-year holds fixed:
 - intended tax rate;
 - public services and public entities;
 - target assessment ratio;
 - local assessment practices.

2.3 Property, transaction, and assessment data

- The main property records come from **ATTOM**, which contains annual observations on roughly **118 million** U.S. properties from **2003–2016**.
- The paper restricts attention to residential properties with up to four units, because commercial and large multifamily properties may be assessed under different rules.

- The baseline sample uses homes with observed arm’s-length, full-consideration transactions, so market value can be measured directly from sales prices.
- Assessment ratios are formed using assessed values and transaction prices observed in the same year.
- California is excluded from the main national sample because Proposition 13 imposes unusually mechanical constraints on assessment growth. The paper reports California evidence separately in the Online Appendix.

2.4 Race and ethnicity data

- Homeowner race and ethnicity come from **Home Mortgage Disclosure Act (HMDA)** records.
- HMDA records include applicant race and ethnicity for mortgage applications and originations.
- The authors merge HMDA to ATTOM using year, census tract, lender name, and mortgage amount.
- The race/ethnicity of the **seller** is the relevant object, because the seller is the owner when the assessment was generated.
- The paper focuses on two main minority groupings:
 - Black mortgage holders;
 - Black or Hispanic mortgage holders.
- The comparison group is non-Hispanic white homeowners.

2.5 Final sample and auxiliary data

- The final baseline data set contains about **6.9 million** homes across **49 states**.
- Each home is linked to:
 - assessed value;
 - transaction price;
 - taxing jurisdiction;
 - homeowner race/ethnicity;
 - census tract and block group;
 - property characteristics;
 - tract-level demographic and economic variables from the American Community Survey.

- For a Cook County case study, the authors merge in administrative assessment-appeal records from **2002–2015**, covering whether a homeowner appealed, whether the appeal succeeded, and the size of the reduction.

2.6 Key descriptive objects

- **Assessment ratio:**

$$AR_{ijt} = \frac{A_{ijt}}{M_{ijt}}, \quad ar_{ijt} = \log(A_{ijt}) - \log(M_{ijt}).$$

- **Assessment gap:** the difference in log assessment ratios between a minority homeowner group and non-Hispanic white homeowners within the same tax jurisdiction and year.

- **Effective tax rate:**

$$ETR_{ijt} = \frac{\text{tax bill}_{ijt}}{M_{ijt}}.$$

- **Spatial component:** the portion of the assessment gap that disappears after conditioning on smaller geographies such as census tracts or block groups.
- **Within-neighborhood component:** the remaining assessment gap after comparing homeowners inside the same tract-year or block-group-year.

3 Models and empirical specifications (all equations + interpretation)

3.1 Tax-equity benchmark

The paper begins from the idea that an equitable proportional property tax requires equal assessment ratios within any taxing jurisdiction.

Let:

- A_{ijt} be assessed value of home i in taxing jurisdiction j and year t ;
- M_{ijt} be market value, measured by transaction price;
- τ_{jt} be the local intended tax rate.

If taxation is proportional to market value and all homes in jurisdiction j face the same intended tax rate, then:

$$\frac{A_{ijt}}{M_{ijt}} = \kappa_{jt}$$

for every home i in jurisdiction j at time t .

Interpretation. The target assessment ratio κ_{jt} can vary across jurisdictions. Some jurisdictions assess homes at 100% of market value; others use target ratios such as 40%, 33.3%, or lower. But within a jurisdiction-year, the ratio should not differ systematically by race, ethnicity, income, or location.

3.2 Baseline assessment-gap specification

The core specification is:

$$\log(A_{ijt}) - \log(M_{ijt}) := ar_{ijt} = \gamma_{jt} + \beta^r \text{Race}_{ijt} + \varepsilon_{ijt}.$$

where:

- ar_{ijt} is the log assessment ratio;
- γ_{jt} is a jurisdiction-year fixed effect;
- Race_{ijt} is a vector of race/ethnicity indicators;
- β^r measures the assessment-ratio difference relative to non-Hispanic white homeowners.

Interpretation. The coefficient β^r is not a price difference and not a difference in public services. It is a within-jurisdiction difference in assessed value relative to market value. A coefficient of 0.10 means the minority group is assessed about 10% higher relative to market value than the white reference group.

3.3 Spatial decomposition with finer geography fixed effects

The authors estimate the same specification using progressively finer geography-year fixed effects:

$$ar_{ijt} = \gamma_{gt} + \beta^r \text{Race}_{ijt} + \varepsilon_{ijt},$$

where g is alternatively:

- taxing jurisdiction;
- jurisdiction $\times$ census tract;
- jurisdiction $\times$ census block group.

Interpretation.

- The jurisdiction-year coefficient measures the full racial assessment gap holding intended tax policy fixed.
- The tract-year or block-group-year coefficient measures the within-neighborhood component.

- The difference between the two captures the spatial component: inequality arising because minority homeowners live in neighborhoods that are overassessed relative to market prices.

3.4 Why not simply control for transaction price?

The paper explains why controlling directly for home price can be a bad control if race affects the formation of market prices through neighborhood sorting, historical discrimination, or public-good capitalization.

Suppose market value is itself a function of race-linked neighborhood forces:

$$M_{ijt} = f(\text{Race}_{ijt}, \Omega_{ijt}),$$

where Ω_{ijt} captures other determinants of price.

Then adding price controls to the assessment-gap regression would recover only a residual racial component rather than total racial inequality. In the paper's notation, adding a price control can yield:

$$ar_{ijt} = \gamma_{jt} + \beta_1^r \text{Race}_{ijt} + \psi (\beta_2^r \text{Race}_{ijt} + \Omega_{ijt}) + \varepsilon_{ijt}.$$

The object of interest is:

$$\beta_1^r + \psi \beta_2^r,$$

not just β_1^r .

Interpretation. If minority neighborhoods have lower market prices because of segregation, disinvestment, or local public-good differences, then controlling for price can remove part of the inequality the paper is trying to measure. The paper therefore separates price-related mechanisms into property attributes and location rather than mechanically controlling for transaction price.

3.5 Property-attribute controls

To test whether the gap is driven by different housing-stock features, the authors estimate:

$$ar_{ijt} = \alpha_{\text{attr}(i)} + \gamma_{jt} + \beta^r \text{Race}_{ijt} + \varepsilon_{ijt}.$$

Here $\alpha_{\text{attr}(i)}$ is a fixed effect for the unique bundle of home attributes, including:

- square footage;
- number of bathrooms;
- home vintage;
- fireplace, patio/porch, pool, and number of stories.

The paper also constructs an attribute-implied home value using leave-state-out hedonic prices and controls for bins of that value.

Interpretation. If minority homeowners are overassessed only because they own homes that are harder to value, the race coefficient should shrink sharply after controlling for detailed property attributes. It does not.

3.6 Transaction-price test

To test whether the assessment gap is mechanically caused by minority homeowners selling at lower prices, the paper studies homes that sell multiple times.

Let P_{i0} be the first observed sale price. A predicted future price is:

$$\hat{P}_{it} = P_{i0} \times HPI_{zt},$$

where HPI_{zt} is ZIP-code-level home price growth.

The regression is:

$$\log(P_{it}) - \log(\hat{P}_{it}) = \gamma_{bg,t} + \beta^r \text{SellerRace}_i + \varepsilon_{izt}.$$

Interpretation. The left-hand side is the unexpected component of transaction price. If Black or Hispanic sellers receive lower prices, this could mechanically raise their assessment ratios. The paper finds the opposite: minority sellers receive a 2%–3% premium relative to predicted prices, so transaction-price differences would, if anything, make the measured assessment gap smaller.

3.7 Hedonic mismatch between market values and assessed values

The mechanism section compares how market values and assessments respond to property and neighborhood characteristics.

For $y \in \{A, M\}$, estimate:

$$\log(y_{injt}) = \gamma_{jt}^y + \beta_{att}^y X_{injt} + \beta_{neigh}^y W_{njt} + \varepsilon_{injt}^y,$$

where:

- X_{injt} are property-level characteristics;
- W_{njt} are tract-level neighborhood characteristics;
- A denotes assessed value;
- M denotes market value.

The key comparison is:

$$\hat{\beta}_{neigh}^M - \hat{\beta}_{neigh}^A.$$

Interpretation. If market prices capitalize neighborhood characteristics more strongly than assessments do, then assessments are too high in negatively valued neighborhoods and too low in positively valued neighborhoods. Because racial residential sorting causes minority homeowners to face different neighborhood characteristics on average, this spatial valuation error generates a racial assessment gap.

3.8 Segregation and spatial sorting

To test whether spatial misvaluation lands more strongly along racial lines in segregated areas, the paper uses the county-level dissimilarity index:

$$dis_C = \frac{1}{2} \sum_{n \in C} \left| \frac{b_n}{B_C} - \frac{w_n}{W_C} \right|,$$

where:

- b_n and w_n are Black and white populations in tract n ;
- B_C and W_C are Black and white populations in county C .

Interpretation. The index is the share of residents who would need to move across tracts to eliminate segregation. If neighborhood misvaluation is important, the racial assessment gap should be larger in more segregated counties. That is what Figure IV shows.

3.9 Appeals mechanism in Cook County

For Cook County, the paper studies appeals using block-group-year comparisons.

For the total effect of appeals on assessment reductions, the paper estimates:

$$\log(\Delta A_{it}) = \gamma_{bt} + \beta^r \text{Race}_{it} + \varepsilon_{it},$$

where ΔA_{it} is the positive reduction in assessed value from appeals.

Interpretation. A negative coefficient means minority homeowners receive smaller assessment reductions from the appeals system, increasing the assessment gap. The paper finds racial differences at every stage: filing, winning, and size of reduction.

4 Identification and instruments (what makes the estimates credible)

4.1 What is identified

- The paper identifies a **tax-equity gap**: racial and ethnic differences in assessed value relative to market value, holding fixed intended tax policy and local public services.
- The baseline estimate should be read as a disparity in tax burden, not as a direct estimate of assessor intent.
- The paper's empirical benchmark is horizontal equity: within a jurisdiction-year, two homes with the same market value should face the same assessed value and thus the same tax burden.

4.2 Why jurisdiction-year fixed effects are central

- Property tax rates and target assessment ratios vary widely across localities.
- Jurisdiction-year fixed effects absorb this variation.
- These fixed effects also ensure that homeowners being compared face the same taxing entities and public services.
- This matters because comparing tax burdens across jurisdictions would mix assessment inequality with real differences in public goods, tax rates, and local fiscal policy.

4.3 Why the result is not driven by property attributes

- A possible concern is that minority homeowners own different types of homes, and those homes are more difficult to assess.
- The paper directly controls for detailed attribute bundles and attribute-implied values.
- The assessment gap remains very similar.
- Within tract and block group, comparing physically similar homes still leaves a large assessment gap.

4.4 Why the result is not driven by lower minority transaction prices

- The assessment ratio has market value in the denominator.
- If Black or Hispanic sellers received lower prices for the same homes, their assessment ratios could mechanically look higher.

- The repeated-sales test shows the opposite: Black sellers receive about **2.2%** more than predicted, and Black or Hispanic sellers about **3.3%** more.
- Therefore transaction-price differences cannot explain the assessment gap; if anything, they imply the baseline gap is understated.

4.5 Why the mechanism evidence is credible

- The spatial evidence has several mutually reinforcing pieces:
 - the gap falls by just over half when moving from jurisdiction-year to block-group-year comparisons;
 - tract minority share predicts higher assessment ratios even controlling for individual homeowner race;
 - assessments underweight neighborhood attributes relative to market prices;
 - the gap is largest in more segregated counties;
 - low-value and low-income minority neighborhoods have especially high assessment ratios.
- The within-neighborhood evidence is supported by administrative appeals data from Cook County.
- The appeals data directly observe a bureaucratic process through which similar homeowners can end up with different assessments over time.

4.6 Limitations

- The paper does not claim to prove overt racial animus by individual assessors.
- It cannot fully decompose all channels behind within-neighborhood inequality; Cook County is a detailed case study, not a national appeals data set.
- Some sample selection arises because the baseline sample requires observed transactions and a successful HMDA match, though the balance tests suggest selection is not large enough to explain the results.
- The analysis excludes California from the main sample because Proposition 13 makes its assessment process mechanically different.

4.7 Relation to the literature

- The paper contributes to work on racial disparities in property taxation by using comprehensive national property-level data rather than only neighborhood aggregates.

- It also contributes to the assessment-regressivity literature by showing a mechanism: neighborhood characteristics are undercapitalized into assessments relative to market prices.
- More broadly, the paper shows how structural inequality can emerge from race-neutral administrative systems when errors correlate with segregated spatial patterns and unequal access to bureaucracy.

5 Tables and figures

5.1 Table I: balance in sample construction

TABLE I
BALANCE TABLE FOR SAMPLE CONSTRUCTION

Tract or property attribute	Observed transaction	Merge to HMDA
Panel A: Property features		
Square feet	-29.85883*** (1.79071)	-10.37782*** (2.96618)
No. baths	0.01367*** (0.00181)	0.0158*** (0.00341)
Year built	0.96571*** (0.23695)	1.7324*** (0.11026)
Patio or porch (binary)	-0.00516*** (0.00060)	0.00906*** (0.00096)
Pool (binary)	-0.00652*** (0.00047)	0.00933*** (0.00071)
Fireplace (binary)	-0.01475*** (0.00082)	0.02363*** (0.00127)
Number of stories	0.00318** (0.00149)	0.03054*** (0.00211)
Panel B: Neighborhood attributes		
Population share Black	-0.00379*** (0.00064)	-0.00657*** (0.00113)
Population share nonwhite	-0.00318*** (0.00063)	-0.00642*** (0.0012)
Population share Black or Hispanic	-0.00503*** (0.00085)	-0.00668*** (0.00136)
Population share white	0.00421*** (0.00078)	0.00616*** (0.00133)
Population (log)	0.00657*** (0.00101)	0.01472*** (0.0014)
Owner percentage	-0.00395*** (0.00045)	0.00703*** (0.0006)
Median age (years)	-0.08956*** (0.01747)	-0.15295*** (0.05312)
Median year purchased	0.28292*** (0.01839)	0.08712*** (0.01941)
Median home value (log)	0.00696*** (0.00156)	0.01448*** (0.00261)
Median HH income (log)	0.00236** (0.00105)	0.02008*** (0.00207)
Unemployment rate	-0.00044*** (0.00011)	-0.00177*** (0.00024)
Not in labor force share	-0.00107*** (0.00023)	-0.00386*** (0.00044)
Gini coefficient	0.00072*** (0.00014)	-0.00247*** (0.0002)
Share SNAP	-0.00119*** (0.00022)	-0.00471*** (0.00059)

TABLE I
CONTINUED

Tract or property attribute	Observed transaction	Merge to HMDA
Panel C: Valuation		
Transaction price (log)	-	0.03896*** (0.00469)
Assessment ratio (log)	-	0.00932*** (0.00185)

Notes. This table reports OLS estimates relating to sample selection on two margins. Column (1) compares the set of observed transactions with a 20% sample of properties that do not transact. Column (2) compares cleaned assessment ratios that can be associated with race and ethnicity (via HMDA) with cleaned assessment ratios that are not matched. For each row, the dependent variable is a dummy variable equal to 1 if an observation enters the relevant subsample, and 0 otherwise. All estimates include jurisdiction-year fixed effects. Errors are clustered at the jurisdiction level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read:

- Table I studies selection on two margins: whether a transaction is observed and whether the transaction can be matched to HMDA race/ethnicity data.
- Homes that enter the transaction sample and HMDA-matched sample differ statistically from excluded homes, but the economic magnitudes are small for most property and neighborhood attributes.

- HMDA-matched homes have transaction prices about **4%** higher and assessment ratios about **1%** higher than unmatched homes.
- The important point is that selection does not generate obvious overrepresentation of high-minority neighborhoods that could explain a 10%–13% assessment gap.

5.2 Table II: baseline assessment gap

Read:

- This is the main baseline table.
- With jurisdiction-year fixed effects, Black homeowners have assessment ratios **12.66%** higher than non-Hispanic white homeowners.
- Black or Hispanic homeowners have assessment ratios **9.84%** higher.
- Moving to jurisdiction-tract-year fixed effects reduces the estimates to **6.40%** and **5.30%**.
- Moving to jurisdiction-block-group-year fixed effects reduces them to **5.88%** and **4.85%**.
- This implies just over half of the assessment gap is spatial and the rest persists within small neighborhoods.

	log(Assessment ratio)		
	(1)	(2)	(3)
Panel A: Black homeowners			
Black mortgage holder	0.1266*** (0.0150)	0.0640*** (0.0020)	0.0588*** (0.0019)
Panel B: Black or Hispanic homeowners			
Black or Hispanic mortgage holder	0.0984*** (0.0106)	0.0530*** (0.0015)	0.0485*** (0.0014)
Fixed effects	Jurisd- year	Jurisd- tract-year	Jurisd-BG- year
No. clusters	37,723	37,723	37,723
Observations	6,987,915	6,987,915	6,987,915

Notes. This table shows our baseline findings of a racial assessment gap. Panel A presents our results for Black homeowners, and Panel B presents our results for Black or Hispanic homeowners. We regress the log assessment ratio on a set of fixed effects at the year $\times$ geography level and on categorical groupings by racial and ethnic identity. Columns (1), (2), and (3) show results using fixed effects at the jurisdiction-year, jurisdiction-tract-year, and jurisdiction-block group-year level, respectively. In all columns, the reference group is non-Hispanic white residents, and for clarity coefficients for groups not being considered in a given column are not reported. The estimates in this table reflect an assessment ratio differential for the given grouping of minority residents relative to non-Hispanic white residents. Standard errors are clustered at the jurisdiction level. * $p < .1$; ** $p < .05$; *** $p < .01$.

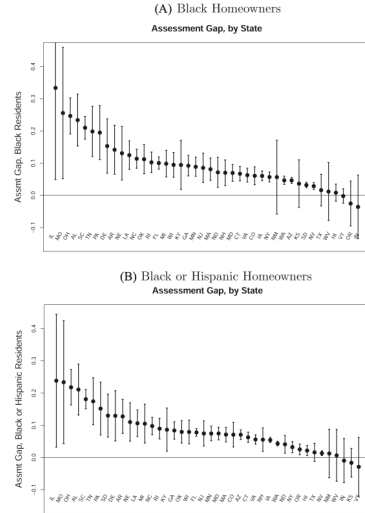


FIGURE I
State-Level Estimates of the Assessment Gap

These graphs show state-level estimates of the assessment gap. For every state with at least 500 observations, we regress log assessment ratio on a jurisdiction-year fixed effect and categorical variables for race and ethnicity. The top graph plots the estimated coefficient for Black mortgage holders, along with a 95% confidence interval. The reference group is non-Hispanic white residents. Standard errors in the underlying regressions are clustered at the jurisdiction level.

5.3 Figure I, Figure II, and Table III: geographic distribution and effective taxes

Read:

- Figure I shows state-level assessment gaps. The gap is positive and statistically significant in most states with sufficient observations.
- Figure II shows the county-level distribution for Black homeowners across **671** counties. The paper reports positive and significant gaps in **391** counties, positive insignificant gaps in **219**, negative insignificant gaps in **53**, and negative significant gaps in only **8**.
- At the 90th percentile of the county distribution, a Black homeowner faces a roughly **27%** higher assessment ratio, equal to about **\$790** annually for a \$207,000 home with a 1.4% tax rate.
- Table III connects assessment ratios to actual tax bills. In the tax-bill subsample, Black homeowners have effective tax rates about **15%** higher; Black or Hispanic homeowners have effective tax rates about **11%** higher.
- The assessment gap is not just an accounting artifact: it translates into higher tax payments.

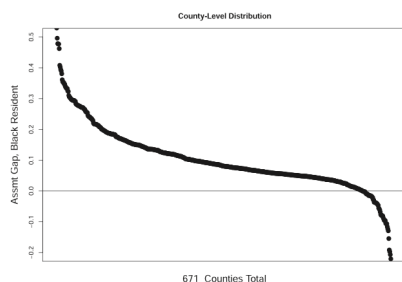


FIGURE II
County-Level Estimates of the Assessment Gap

This graph shows county-level estimates of the assessment gap for Black residents. For every county with at least 500 observations, we regress log assessment ratio on a jurisdiction-year fixed effect and categorical variables for race and ethnicity. We have sufficient data in 671 counties. We plot the estimated coefficient. For visual clarity, we do not include confidence intervals. Point estimates are positive and significant at 5% in 391 counties, positive and insignificant in 219 counties, negative and insignificant in 53 counties, and negative and significant at 5% in 8 counties. The reference group is non-Hispanic white residents. Standard errors in the underlying regressions are clustered at the jurisdiction level.

	Assmt. gap (1)	Before exemptions (2)	Tax bill (3)	Assmt. gap (4)	Before exemptions (5)	Tax bill (6)
Black mortgage holder	0.1290*** (0.0170)	0.1466*** (0.0186)	0.1502*** (0.0182)			
Black or Hispanic mortgage holder				0.0971*** (0.0125)	0.1111*** (0.0120)	0.1145*** (0.0116)
Jurisdiction-year FE	Y	Y	Y	Y	Y	Y
No. clusters	34,776	34,776	34,776	34,776	34,776	34,776
Observations	5,574,777	5,574,777	5,574,777	5,574,777	5,574,777	5,574,777
R ²	0.8857	0.6755	0.6405	0.8856	0.6753	0.6403

Notes: This table shows that inequality in assessment ratios aligns closely with inequality in effective tax rates. Columns (1) and (4) estimate the assessment gap within the sample of homes for which we observe tax payments. Columns (2) and (5) repeat the baseline estimation, but use the effective tax rate (gross of exemptions) as the dependent variable instead of the assessment ratio. Columns (3) and (6) use the effective tax rate after subtracting reported exemptions from the gross tax bill. Effective tax rates are computed using the observed market value of the home. We trim any observation above a calculated effective tax rate of 20% both before and out of exemptions. We believe this to be a conservative choice as 20% is far higher than any property tax rate of which we are aware (the national median is approximately 1.4%), and is likely to be a data error. All specifications use jurisdiction-year fixed effects to hold constant the level of intended taxation. Standard errors are clustered at the jurisdiction level. * $p < .1$, ** $p < .05$, *** $p < .01$.

5.4 Table IV and Table V: what does not explain the gap

Read:

- Table IV asks whether the gap is driven by housing-stock characteristics.
- In the subsample with attributes, the baseline gap remains about **12.03%** for Black homeowners and **9.33%** for Black or Hispanic homeowners.

	log(Assessment ratio)						
	Baseline	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Attributes and jurisdiction (not interacted)							
Black mortgage holder	0.1201*** (0.0082)	0.1189*** (0.0081)		0.1201*** (0.0082)		0.1201*** (0.0082)	
Black or Hispanic mortgage holder	0.0920*** (0.0056)		0.0915*** (0.0055)		0.0920*** (0.0056)		0.0920*** (0.0056)
Panel B: Attributes $\times$ jurisdiction							
Black mortgage holder	0.1201*** (0.0082)	0.1092*** (0.0081)		0.1195*** (0.0087)		0.1218*** (0.0093)	
Black or Hispanic mortgage holder	0.0920*** (0.0056)		0.0852*** (0.0053)		0.0910*** (0.0060)		0.0921*** (0.0065)
Panel C: Attributes $\times$ tract							
Black mortgage holder	0.0675*** (0.0022)	0.0562*** (0.0020)		0.0602*** (0.0023)		0.0553*** (0.0023)	
Black or Hispanic mortgage holder	0.0559*** (0.0016)		0.0463*** (0.0015)		0.0494*** (0.0017)		0.0454*** (0.0017)

TABLE IV CONTINUED							
log(Assessment ratio)							
	Baseline	(1)	(2)	(3)	(4)	(5)	(6)
Panel D: Attributes $\times$ block group							
Black mortgage holder	0.0614*** (0.0021)	0.0484*** (0.0018)		0.0530*** (0.0021)		0.0475*** (0.0020)	
Black or Hispanic mortgage holder	0.0510*** (0.0015)		0.0409*** (0.0013)		0.0440*** (0.0015)		0.0400*** (0.0016)
Price FE							
No. clusters	Baseline	Att. Bin	Baseline	Att. Bin	200Q	500Q	500Q
Observations	25,798	25,798	25,798	25,798	25,798	25,798	25,798
	4,674,430	4,674,430	4,674,430	4,674,430	4,674,430	4,674,430	4,674,430

Notes: This table shows our baseline findings of a racial assessment gap controlling for attributes of the property and attribute-implied home value. Panel A presents results controlling for attributes of the property and attribute-implied home value without interacting these with geographic fixed effects. Panel B controls for attributes of the property and attribute-implied home value interacted with jurisdiction-year fixed effects. Panel C controls for attributes of the property and attribute-implied home value interacted with tract-year fixed effects. Panel D controls for attributes of the property and attribute-implied home value interacted with block group-year fixed effects. In all specifications, we regress the log assessment ratio on geography-year fixed effects and on categorical groupings by racial and ethnic identity. Baseline estimates are presented on the left for ease of interpretation. Columns (1) and (2) control for attributes using attribute fixed effects. Columns (3) and (4) use 200 attribute-implied home value bins as fixed effects. Columns (5), (6), and (7) present results for Black homeowners only. Columns (2), (4), and (6) present results for Black and Hispanic homeowners. In all columns, the reference group is non-Hispanic white residents, and for clarity coefficients for groups not being considered in a given column are not reported. The estimates in this table reflect an assessment ratio differential for the given grouping of minority residents relative to non-Hispanic white residents. Standard errors are clustered at the jurisdiction level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Implied Elasticity of Assessment Ratio to One Standard Deviation Shift

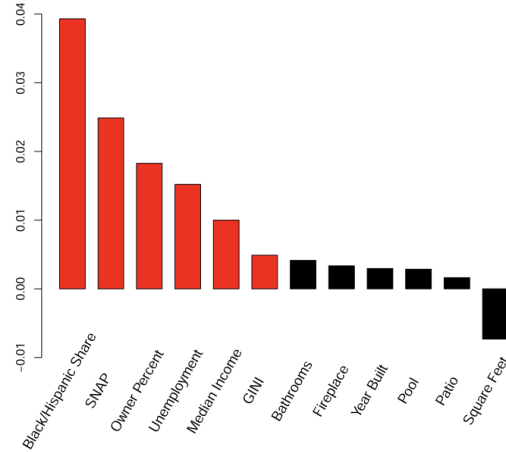


FIGURE III

Hedonic Models: Mismatch

Each bar in this figure plots the difference between two estimated hedonic prices: one estimated from a model with market values as the dependent variable, and one from a model with assessment values as the dependent variable. Otherwise, the hedonic models are identical: all regressors are the same. Both market values and assessed values are logged in the underlying models, so the difference between the estimated hedonic prices represents a proportional shift in the assessment ratio that arises from a one standard deviation shift in the underlying variable. Bars in light gray/red are tract-level characteristics. Bars in black are property-level characteristics. A bar at zero would denote that the market hedonic price is the same as the assessment hedonic price. Larger bars signify a greater disconnect between market hedonics and assessment hedonics. Finally, bars above zero denote that estimated market hedonic prices are greater in (absolute) magnitude than assessed hedonic prices. Bars below zero denote that the assessment hedonic price is larger. [Online Appendix Table A5](#) shows the estimated prices that underlie this figure.

- Adding detailed attribute fixed effects or attribute-implied price bins barely changes the estimates.
- Even within tract or block group, comparing physically similar homes leaves economically meaningful gaps of roughly **4%–6%**.
- Table V asks whether minority sellers receive lower transaction prices. They do not: Black sellers receive about **2.2%** more than predicted, and Black or Hispanic sellers about **3.3%** more.
- Therefore the measured assessment gap is not caused by minority homeowners selling at low prices.

5.5 Table VI and Figure III: neighborhood misvaluation

Read:

- Table VI adds tract-level demographic shares to the baseline regression.
- Individual race still matters, but neighborhood racial composition also strongly predicts higher assessment ratios.
- The coefficient on tract Black share is about **0.299**; the coefficient on tract Black or Hispanic share is about **0.277**.
- Figure III compares hedonic prices implied by market values and by assessments.
- Assessments line up closely with market prices for most property-level attributes; the property-attribute mismatches are generally below **1%** for a one-standard-deviation shift.
- Assessment models are much less aligned with market prices for neighborhood characteristics.
- Economic message: assessors do not ignore neighborhoods, but they underweight neighborhood characteristics relative to markets.

5.6 Figures IV–VII: segregation, home values, and income

Read:

- Figure IV sorts counties by racial segregation. The assessment gap rises with segregation and is especially high in the most segregated decile.
- Figure V sorts tracts by median home value. Above the median, the gap is relatively stable; in the lower half of the home-value distribution, the gap increases sharply and exceeds **15%** in the lowest vigintile.
- Figure VI double-sorts neighborhoods by home value and minority share. Holding neighborhood value group fixed, excess assessment rises with minority share.
- Figure VII sorts tracts by median income. Above-median-income tracts show a roughly **5%** gap; low-income minority neighborhoods have much larger gaps.
- These figures show that the assessment gap is not simply about low home values or low income. Minority composition matters within those bins.

5.7 Figure VIII and Table VII: within-neighborhood inequality and appeals

Read:

- Figure VIII shows within-block-group assessment gaps by homeowner income quantile.

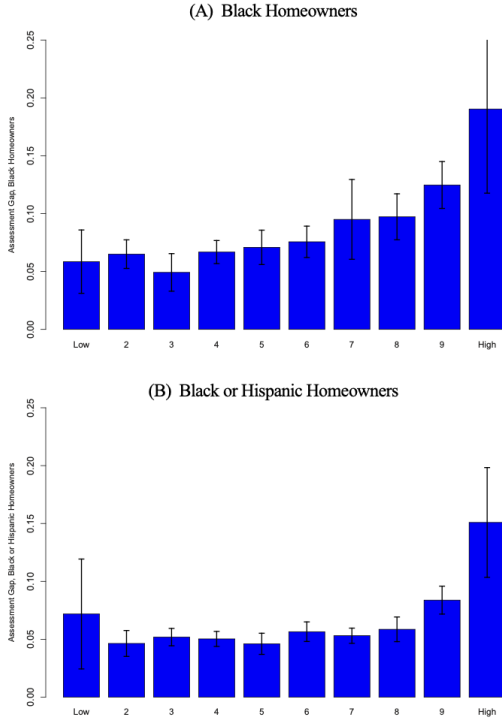


FIGURE IV

Assessment Gap by Racial Segregation Decile

In each panel, we assign counties to deciles by a county-level segregation measure, constructed from tract-level demographics from the 2000 decennial census. It is a stylized fact that larger counties have more segregation. As a result, the lowest deciles have substantially fewer observations than higher deciles. We estimate inequality separately in each decile following equation (1). Full regression output is available in the [Online Appendix](#).

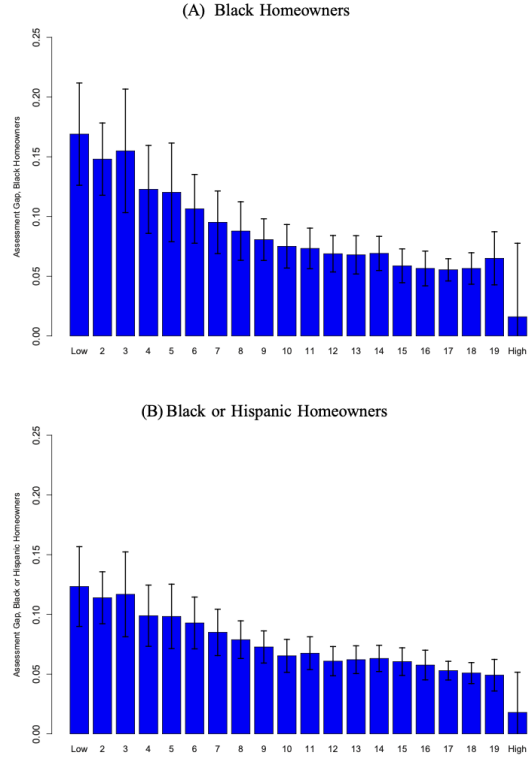


FIGURE V

Assessment Gap by Tract Median Home Value Vigintile

This figure presents tract-level average assessment gaps by neighborhood-level home values. In each panel, we assign tracts to 20 quantiles based on the tract-level distribution of median home value.

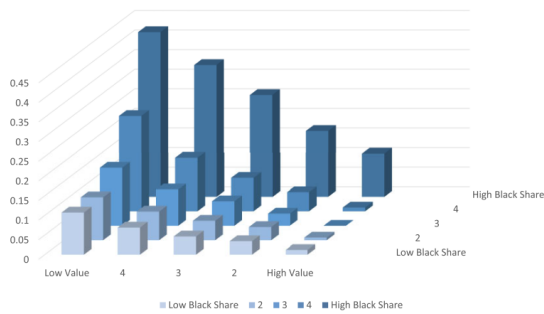
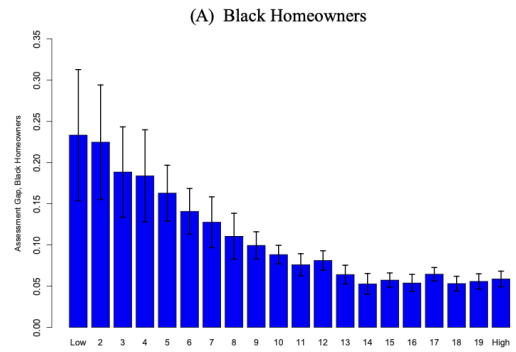


FIGURE VI

Average "Excess" Assessment by Demographics and Neighborhood Home Value

This figure splits neighborhoods into quantiles based on median home value using tract-level measures from the ACS and, in each quantile, splits homes by neighborhood demographic share. Using a pooled regression, we estimate "excess" assessment—that is, assessment ratio deviation from taxing jurisdiction-year average—for each bin.



(B) Black or Hispanic Homeowners

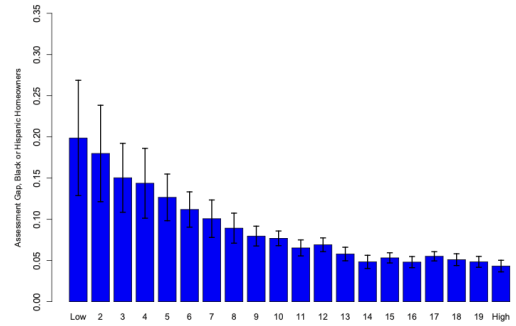


FIGURE VII

Assessment Gap by Tract Income Quantile

This figure presents tract-level average assessment gaps by neighborhood-level income. In each panel, we assign tracts to each of 20 quantiles based on the tract-level distribution of median income.

- The gap remains around **5%–6%** across much of the income distribution and is largest in the highest-income quintile.
- Table VII studies Cook County appeals.
- Black homeowners are about **1.08 percentage points** less likely to appeal, **2.24 percentage points** less likely to win conditional on appealing, and receive reductions about **0.48 percentage points** smaller conditional on success.
- For Black or Hispanic homeowners, the corresponding differences are **1.16 percentage points**, **2.05 percentage points**, and **0.26 percentage points**.
- One appeal cycle increases inequality by about **20 basis points** for Black homeowners and **16 basis points** for Black or Hispanic homeowners.
- Since appeals can occur repeatedly over a homeowner's tenure, the cumulative effect can explain a meaningful portion of within-neighborhood inequality.

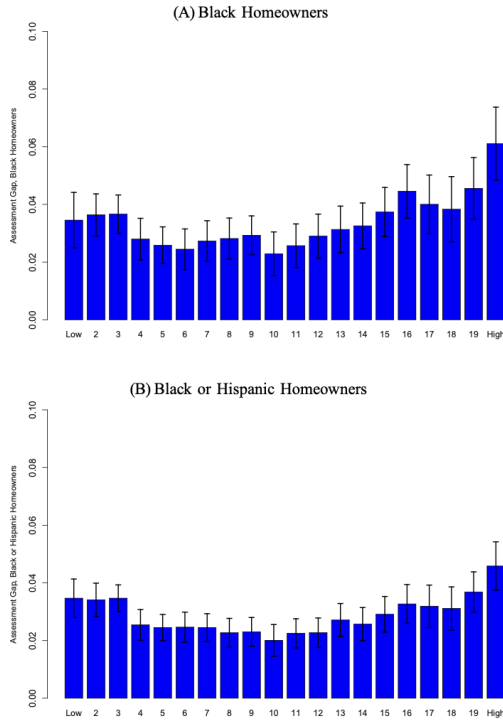


FIGURE VIII

Assessment Gap by Homeowner Income Quintile

In each panel, we assign tracts to 20 quantiles based on homeowner reported income in HMDA records.

	Dependent variable:				
	Assessment ratio/BG (%) (1)	Appeal (2)	Win appeal (3)	Reduction (4)	Total effect (5)
Panel A: Black homeowners					
Black mortgage holder	5.231*** (0.585)	-1.075*** (0.104)	-2.243*** (0.368)	-0.478*** (0.119)	-0.202*** (0.021)
Panel B: Black or Hispanic homeowners					
Black or Hispanic mortgage holder	5.118*** (0.426)	-1.158*** (0.080)	-2.054*** (0.254)	-0.259*** (0.075)	-0.161*** (0.014)
Baseline rate	NA	14.6	67.4	12.0	N/A
Fixed effects	BG-year	BG-year	BG-year	BG-year	BG-year
No. clusters	426	3,954	3,924	3,981	3,954
Observations	141,535	3,072,521	617,157	441,424	3,071,538

Notes: This table uses administrative microdata on property tax appeals in Cook County. Column (1) shows the baseline within-block group assessment gap in Cook County. Column (2) shows unconditional propensity to appeal. Column (3) conditions on a homeowner having filed an assessment appeal. Column (4) conditions on a successful appeal. Column (5) estimates the total effect of appeals on inequality within tax year. In columns (2) and (3), the dependent variable is a binary indicator. In column (4), the dependent variable is the reduction amount divided by the proposed assessment. In column (5), the dependent variable is the log difference between preappeal and postappeal assessments. Homeowners who do not appeal are assumed to have zero change. Fixed effects across all columns are at the block-group-year level. Standard errors are clustered at the block-group level. The baseline rates for (i) appeal propensity, (ii) winning appeal, and (iii) reduction conditional on a successful appeal are reported in the first line below the estimates. Coefficients, standard errors, and baseline rates are reported as percent. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.8 Table VIII: assessment gap by year

Read:

TABLE VIII
ASSESSMENT GAP BY YEAR

	log(Assessment ratio)											
	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Black homeowners												
Black mortgage holder	0.0309*** (0.0082)	0.0354*** (0.0084)	0.0844*** (0.0111)	0.1765*** (0.0217)	0.1914*** (0.0334)	0.1701*** (0.0339)	0.1628*** (0.0244)	0.1822*** (0.0283)	0.1497*** (0.0116)	0.1691*** (0.0316)	0.1254*** (0.0089)	0.1077*** (0.0100)
Panel B: Black or Hispanic homeowners												
Black or Hispanic mortgage holder	-0.0001 (0.0075)	0.0190*** (0.0068)	0.0596*** (0.0086)	0.1526*** (0.0150)	0.1637*** (0.0194)	0.1334*** (0.0224)	0.1293*** (0.0182)	0.1375*** (0.0197)	0.1093*** (0.0080)	0.1208*** (0.0221)	0.0842*** (0.0056)	0.0724*** (0.0064)
Fixed effects	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year	Jurisd- year
No. clusters	14,683	15,739	16,563	15,456	16,457	17,749	18,177	18,963	18,719	19,756	24,269	15,888
Observations	666,184	609,361	579,293	489,501	524,133	473,830	502,070	522,700	584,978	561,824	820,940	648,098

Notes: This table shows our findings of a racial assessment gap by year. Panel A presents our results for Black homeowners, and Panel B presents our results for Black or Hispanic homeowners. In all specifications, we regress the log assessment ratio on jurisdiction-year fixed effects and on categorical groupings by racial and ethnic identity. In all columns, the reference group is non-Hispanic white residents, and for clarity coefficients for groups not being considered in a given column are not reported. The estimates in this table reflect an assessment ratio differential for the given grouping of minority residents relative to non-Hispanic white residents. Standard errors are clustered at the jurisdiction level. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Table VIII shows the gap year by year from **2005–2016**.
- The gap rises sharply around the financial crisis: for Black homeowners, it increases from about **3%** in 2005 to about **17.7%** in 2008 and **19.1%** in 2009.
- For Black or Hispanic homeowners, the estimate is approximately zero in 2005, then rises to **15.3%** in 2008 and **16.4%** in 2009.
- The gap remains elevated for years after the housing crash and only partially declines by 2016.
- This pattern is consistent with minority neighborhoods experiencing larger price declines while assessments adjust slowly or imperfectly.

6 Short summary

- **Method.** The paper measures racial inequality in property taxation by comparing log assessment ratios within taxing jurisdiction-year cells.
- **Data.** It links national property assessments and transactions from ATTOM to homeowner race/ethnicity from HMDA, tax-jurisdiction geography from GIS shapefiles, and neighborhood characteristics from ACS data.
- **Main result.** Black homeowners face assessment ratios about **12.7%** higher than white homeowners; Black or Hispanic homeowners face assessment ratios about **9.8%** higher.
- **Mechanism.** Just over half of the gap is spatial: assessments are less responsive to neighborhood characteristics than market prices, and segregation causes these errors to fall disproportionately on minority homeowners.
- **Within-neighborhood result.** The remaining gap persists within block groups. Cook County evidence suggests that differential use and success of assessment appeals can contribute to this component.
- **Policy implication.** Improving assessment models and reducing disparities in appeals access could reduce racial inequality in property tax burdens without changing statutory tax rates.

7 Conclusion

7.1 Core conclusion

- The paper documents a large, nationwide racial assessment gap in property taxation.
- The gap means that minority homeowners pay higher effective tax rates than white homeowners in the same jurisdictions for the same bundle of public services.
- The inequality is not explained by lower transaction prices, housing-stock differences, reassessment frequency, or assessment growth caps.

7.2 Economic intuition

- A property tax can be legally proportional yet administratively unequal if assessed values do not track market values equally across space and households.
- Market prices capitalize neighborhood characteristics sharply; assessments capitalize them less precisely.
- Because U.S. neighborhoods are racially segregated, spatial valuation mistakes become racialized tax burdens even without overt discriminatory intent.
- Appeals can then amplify inequality if white homeowners are more likely to use the appeals system or receive larger reductions.

7.3 Takeaway

This paper is important because it shows that racial inequality can arise inside a seemingly technical administrative system. The property tax is formally local and race-neutral, but imperfect assessments and unequal bureaucratic engagement produce systematically higher tax burdens for Black and Hispanic homeowners. The central lesson is that tax equity requires not only equal statutory rates, but also accurate and equitable administrative valuation.